INCH POUND

MIL-C-3098/97E
5 May 1993
SUPERSEDING
MIL-C-3098/97D
8 March 1983

MILITARY SPECIFICATION SHEET

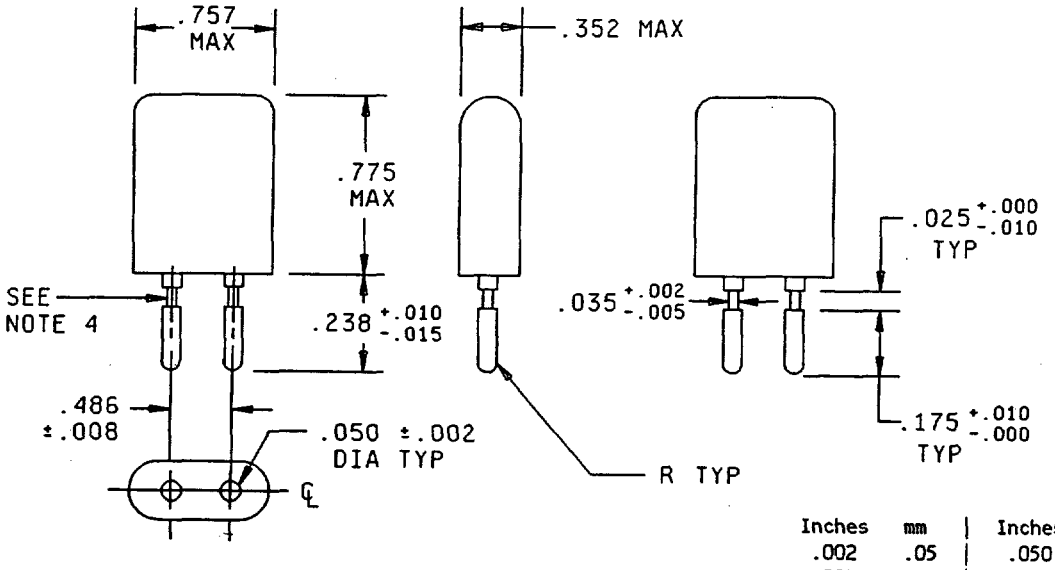
CRYSTAL UNIT, QUARTZ, CR120/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

INACTIVE FOR NEW DESIGN

Pertinent characteristics: 2 to 10 MHz; fundamental; controlled; antiresonance.



Inches	mm	Inches	mm
.002	.05	.050	1.27
.005	.13	.175	4.45
.008	.20	.238	6.05
.010	.25	.352	8.94
.015	.38	.486	12.34
.025	.64	.757	19.23
.035	.89	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR120/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-27/U (MIL-H-10056/14).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2 to 10 MHz, inclusive.

Frequency tolerance: ±5 ppm at +85°C ±5°C.

Antiresonance load capacitance: 100 ±0.5 pF.

Mode of oscillation: Fundamental.

Temperature range:

Operating: +85°C ±5°C.

Operable: -55°C to +90°C.

Drive level, rated: 2 µW.

Crystal quality (Q): 1/

2 to 5 MHz, 1×10^6 minimum.

5+ to 10 MHz, $.5 \times 10^6$ minimum.

Frequency temperature coefficient:

Turning point in the range of: +85°C ±5°C.

Turning point to be marked on holder: To the nearest 0.1°C.

Power-frequency coefficient: ±0.01 ppm maximum from 1 to 3 µW.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging: 2/

2 to 5 MHz, 0.01 ppm per week for the last 2 weeks of a 4-week aging test. 5+ to 10 MHz, 0.02 ppm per week for the last 2 weeks of a 4-week aging test.

1/ $Q = \dfrac{1}{4\pi(\Delta f)(C_T)(R)}$

Where: Δf = difference, in Hz, between the series and antiresonant frequency at C_L (load capacitance) of 100 pF.
C_T = Capacitance: C_o (shunt capacitance of the crystal unit) plus C_L (load capacitance) of 100 pF.
R = Motional-arm resistance of the crystal unit.

2/ The crystal units shall be preconditioned by storage in an oven at 85°C ±5°C for 3 weeks at the time of manufacture.

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Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 204 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Low temperature storage:

Resistance change permitted: ±15 percent.

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

NOTE: NATO guidance type NXT 46.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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